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Question 1: Linear Model (No Preferences)

Problem overview. The problem we aim to solve is a TA assignment and cost minimization problem. We want to decide how many BUs to assign to each TA applicant for each math course, such that all course BU needs are met, all applicant BU requests are satisfied, and the total cost (including prep BUs) is minimized.

Model: Parameters.

Applicants. There are 20 applicants, denoted $a_i \in \mathcal{A} = \{a_1, \ldots, a_{20}\}$. Each applicant belongs to one of two programs:

- Ph.D. students, paid at wage $w_{\text{PhD}} = \$1{,}662$ per BU.
- MSc students, paid at wage $w_{\text{MSc}} = \$1{,}432$ per BU.

We write w_i for the wage per BU of applicant i, where $w_i = w_{\text{PhD}}$ or $w_i = w_{\text{MSc}}$ depending on their program.

Positions. There are 9 open positions, partitioned by appointment type:

$$\begin{aligned}
\mathcal{J} &= \{\text{MATH150, MATH254, MACM316, MATH260, MATH338, MATH342, MATH426, MATHACW, MATHC}\\
\mathcal{W} &= \{\text{MATHACW, MATHCW}\} \quad \text{(workshops)}\\
\mathcal{T} &= \{\text{MATH254, MACM316, MATH260, MATH342}\} \quad \text{(tutorials)}\\
\mathcal{S} &= \{\text{MATH150}\} \quad \text{(seminars)}\\
\mathcal{M} &= \{\text{MATH338, MATH426}\} \quad \text{(marking)}
\end{aligned}$$

We also define $\mathcal{J}_{\text{TS}} = \mathcal{T} \cup \mathcal{S}$ as the set of tutorial and seminar positions.

BU parameters. Each appointment type has a BU step size s_j restricting how many BUs can be assigned:

- s_j = any real value, for workshops ($j \in \mathcal{W}$).
- $s_j = 0.5$, for marking and seminars ($j \in \mathcal{M} \cup \mathcal{S}$).
- $s_j = 1$, for tutorials ($j \in \mathcal{T}$).

Each position $j \in \mathcal{J}$ has a minimum and maximum total non-prep BU need:

$$b_j^{\min} \;\leq\; \text{total BUs at position } j \;\leq\; b_j^{\max}.$$

Course	Type	Min BUs ($b_j^{\min}$)	Max BUs ($b_j^{\max}$)
MATH150	SEM	1.5	2.0
MATH254	TUT	3.0	3.0
MACM316	TUT	9.0	9.0
MATH260	TUT	6.0	7.0
MATH338	MRK	0.5	1.5
MATH342	TUT	3.0	3.0
MATH426	MRK	0.5	1.5
MATHACW	WKP	20	25
MATHCW	WKP	12	16

Each applicant $i \in \mathcal{A}$ requests a minimum and maximum number of non-prep BUs:

$$\ell_i \;\leq\; \text{total non-prep BUs worked by } i \;\leq\; u_i.$$

On top of worked BUs, each applicant i is paid prep BUs p_{ij} for each position j they are assigned to. Prep BUs depend on both the applicant and the appointment type:

- p_{ij}^{WKP}: prep BUs for workshop appointments (varies by applicant).
- p_{ij}^{TUT}: prep BUs for tutorial appointments (varies by applicant).
- p_{ij}^{SEM}: prep BUs for seminar appointments (varies by applicant).
- $p_{ij}^{\text{MRK}} = 0$: no prep BUs for marking.

Prep BUs are paid once per position assigned and are not counted toward the applicant's BU request range. We write p_{ij} for the prep BUs of applicant i at position j, where the value is looked up from the data by appointment type.

For reference, the large constant used in big-M constraints is set to $M = \max_i u_i$.

Model: Decision Variables.

- $x_{ij} \geq 0$: non-prep BUs assigned to applicant i at position j (continuous for WKP; otherwise constrained by step size).
- $y_{ij} \in \{0, 1\}$: equals 1 if applicant i is assigned to position j, and 0 otherwise.
- $n_{ij} \in \mathbb{Z}_{\geq 0}$: number of BU steps assigned to applicant i at position j, for non-workshop positions only, so that $x_{ij} = s_j \cdot n_{ij}$.

Model: Objective Function. Minimize total cost including both worked BUs and prep BUs, weighted by each applicant's wage:

$$\min \quad \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i \big(x_{ij} + p_{ij} \cdot y_{ij}\big).$$

Constraints.

1. **Position BU needs are satisfied.** For each position $j \in \mathcal{J}$:

$$b_j^{\min} \;\le\; \sum_{i \in \mathcal{A}} x_{ij} \;\le\; b_j^{\max}.$$

2. **Applicant BU requests are satisfied.** For each applicant $i \in \mathcal{A}$:

$$\ell_i \;\le\; \sum_{j \in \mathcal{J}} x_{ij} \;\le\; u_i.$$

3. **BU step consistency.** For non-workshop positions $j \in \mathcal{J} \setminus \mathcal{W}$:

$$x_{ij} = s_j \cdot n_{ij}, \quad n_{ij} \in \mathbb{Z}_{\ge 0} \qquad \forall\, i \in \mathcal{A}.$$

   For all positions, the big-M linking constraints ensure $x_{ij} = 0$ when $y_{ij} = 0$, and at least one step when $y_{ij} = 1$:

$$x_{ij} \le M \cdot y_{ij} \qquad \forall\, i \in \mathcal{A},\; j \in \mathcal{J},$$
$$x_{ij} \ge s_j \cdot y_{ij} \qquad \forall\, i \in \mathcal{A},\; j \in \mathcal{J} \setminus \mathcal{W}.$$

4. **At most one workshop per applicant.**

$$\sum_{j \in \mathcal{W}} y_{ij} \le 1 \qquad \forall\, i \in \mathcal{A}.$$

5. **At most one tutorial or seminar per applicant.**

$$\sum_{j \in \mathcal{J}_{\mathrm{TS}}} y_{ij} \le 1 \qquad \forall\, i \in \mathcal{A}.$$

6. **At most three positions per applicant.**

$$\sum_{j \in \mathcal{J}} y_{ij} \le 3 \qquad \forall\, i \in \mathcal{A}.$$

Model: Complete Formulation.

$$
\begin{aligned}
\min \quad & \sum_{i\in\mathcal{A}}\sum_{j\in\mathcal{J}} w_i\big(x_{ij}+p_{ij}\,y_{ij}\big) & \\
\text{subject to} \quad & b_j^{\min} \le \sum_{i\in\mathcal{A}} x_{ij} \le b_j^{\max} & \forall\, j\in\mathcal{J} \\
& \ell_i \le \sum_{j\in\mathcal{J}} x_{ij} \le u_i & \forall\, i\in\mathcal{A} \\
& x_{ij} = s_j\, n_{ij},\ n_{ij}\in\mathbb{Z}_{\ge 0} & \forall\, i\in\mathcal{A},\ j\in\mathcal{J}\setminus\mathcal{W} \\
& x_{ij} \le M\, y_{ij} & \forall\, i\in\mathcal{A},\ j\in\mathcal{J} \\
& x_{ij} \ge s_j\, y_{ij} & \forall\, i\in\mathcal{A},\ j\in\mathcal{J}\setminus\mathcal{W} \\
& \sum_{j\in\mathcal{W}} y_{ij} \le 1 & \forall\, i\in\mathcal{A} \\
& \sum_{j\in\mathcal{J}_{\mathrm{TS}}} y_{ij} \le 1 & \forall\, i\in\mathcal{A} \\
& \sum_{j\in\mathcal{J}} y_{ij} \le 3 & \forall\, i\in\mathcal{A} \\
& x_{ij} \ge 0 & \forall\, i\in\mathcal{A},\ j\in\mathcal{J} \\
& y_{ij} \in \{0,1\} & \forall\, i\in\mathcal{A},\ j\in\mathcal{J}.
\end{aligned}
$$

This is a Mixed Binary Integer Linear Program (MBILP) with $|\mathcal{A}|\cdot|\mathcal{J}| = 20\times 9 = 180$ binary variables y_{ij}, up to 180 continuous/integer variables x_{ij}, and up to 180 integer variables n_{ij} for non-workshop positions.

Question 2: Augmented Model with Applicant Preferences

Overview. Each applicant $i \in \mathcal{A}$ submits an ordered preference list of up to four positions. To account for preferences in a simple and tractable way, we focus only on a fraction of applicant getting **any of their 4 choice**. Let $j_i^* \in \mathcal{J}$ denote thefirst-choice position of applicant i. Our goal is to ensure that a minimum fraction of applicants receive any of their 4 choices appointment, while still minimizing total cost.

This approach is simpler and more transparent than a full penalty-based method: it avoids the need to calibrate a penalty weight λ, and the coverage parameter α has a direct and intuitive interpretation.

New Variable. Introduce a binary indicator $z_i \in \{0, 1\}$ for each applicant $i \in \mathcal{A}$, where $z_i = 1$ if applicant i is assigned to any of their preferred positions, and $z_i = 0$ otherwise.

New Constraints.

1. **Link z_i to any of their preferred assignment.** The variable z_i can equal 1 only if applicant i is actually assigned to their first-choice position j_i^*:

$$z_i \leq \sum_{k=1}^{4} y_{i,j_i^k} \qquad \forall\, i \in \mathcal{A}.$$

2. **Minimum choice coverage.** At least a fraction $\alpha \in [0, 1]$ of all applicants must receive their first-choice appointment:

$$\sum_{i \in \mathcal{A}} z_i \;\geq\; \alpha\, |\mathcal{A}|.$$

   For example, setting $\alpha = 0.5$ requires that at least 10 out of 20 applicants receive their first-choice position.

3. **Integrality.**

$$z_i \in \{0, 1\} \qquad \forall\, i \in \mathcal{A}.$$

Objective Function. The objective from Question 1 is unchanged:

$$\min \quad \sum_{i \in \mathcal{A}} \sum_{j \in \mathcal{J}} w_i \big(x_{ij} + p_{ij}\, y_{ij} \big).$$

Preferences enter only through the new constraints, not the objective. This keeps the cost minimization goal clear and avoids introducing an arbitrary penalty weight.

Complete Augmented Formulation.

$$
\begin{array}{rll}
\min & \displaystyle\sum_{i\in\mathcal{A}}\sum_{j\in\mathcal{J}} w_i\big(x_{ij} + p_{ij}\, y_{ij}\big) & \\
\text{subject to} & \text{all constraints from Question 1,} & \\
& z_i \le \displaystyle\sum_{k=1}^{4} y_{i,j_i^k} & \forall\, i \in \mathcal{A} \\
& \displaystyle\sum_{i\in\mathcal{A}} z_i \ge \alpha\,|\mathcal{A}| & (\alpha = 0.5) \\
& z_i \in \{0,1\} & \forall\, i \in \mathcal{A}.
\end{array}
$$

This adds only $|\mathcal{A}| = 20$ binary variables z_i and $|\mathcal{A}| + 1 = 21$ constraints to the model from Question 1, keeping the overall structure as an MBILP.

Discussion. Focusing on first choices only has three practical advantages:

- **Simplicity.** The model adds only one new parameter α whose meaning is immediately clear: the minimum fraction of applicants who get their top pick.
- **No calibration needed.** A penalty-based approach would require choosing a weight λ to balance monetary cost against preference satisfaction, with no natural scale. Our constraint-based approach avoids this entirely.
- **Feasibility control.** If α is set too high the model may become infeasible (e.g. two applicants share the same first choice but the position can only absorb one of them). Lowering α relaxes the constraint until a feasible solution exists. A reasonable starting value is $\alpha = 0.5$, meaning at least half of applicants receive their first choice.